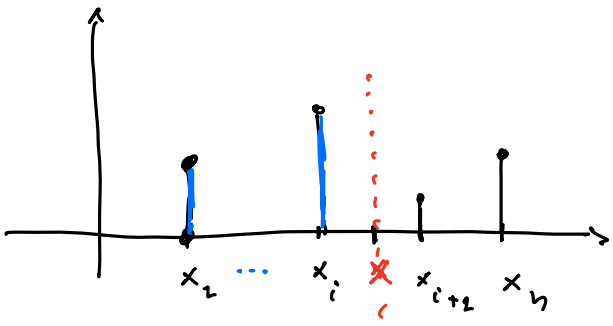
Lezione 08/04/26 3 ore

Funzione di ripartizione o di distribuzione cumulata

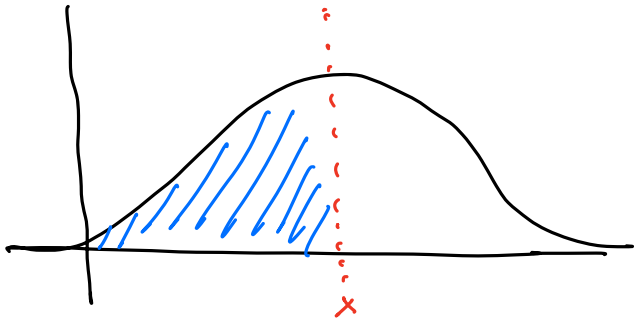
$X : \Omega \to \mathbb{R}$ variabile aleatoria

$F_X : \mathbb{R} \to \mathbb{R}$ $\quad F_X(x) = \mathbb{P}[X \leq x]$



caso discreto

$$F_X(x) = \sum_{x_k \leq x} p_k$$



caso con densità

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt$$

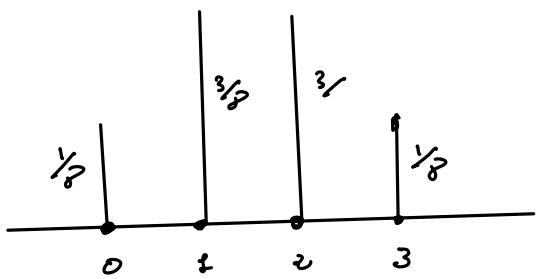
funzione integrale

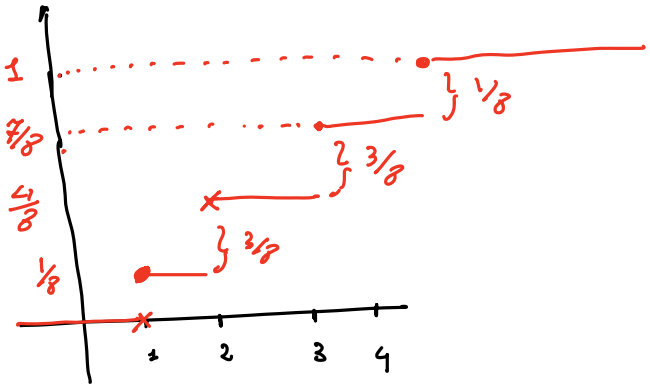
Esempio:

Lancio una moneta equilibrata 3 volte

X = numero di teste

x_i	$\mathbb{P}[X = x_i]$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$





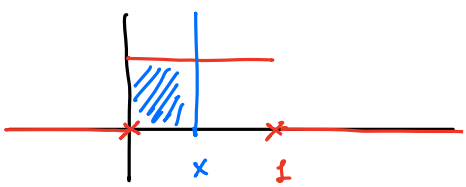
$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1/8 & 0 \le x < 1 \\ 4/8 & 1 \le x < 2 \\ 7/8 & 2 \le x < 3 \\ 1 & x \ge 1 \end{cases}$$

x_i	$\mathbb{P}[X = x_i]$	$F_X(x_i)$
0	1/8	1/8
1	3/8	4/8
2	3/8	7/8
3	1/8	1

Esempio

X variabile con densità f

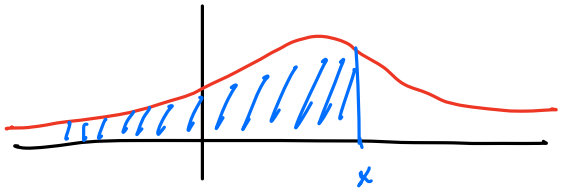
$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{altrove} \end{cases}$$



$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt = \text{AREA BLU} = \begin{cases} 0 & x < 0 \\ x & 0 \le x \le 1 \\ 1 & x \ge 1 \end{cases}$$

Proprietà

X variabile aleatoria $\qquad F_X(x) = \mathbb{P}[X \le x]$ = AREA BLU



a) $F_X(x_1) \le F_X(x_2)$ se $x_1 \le x_2$

b) $\lim_{x \to -\infty} F_X(x) = 0 \qquad \lim_{x \to +\infty} F_X(x) = 1$

c) $\lim_{x \to x_0^+} F_X(x) = F_X(x_0)$ $\qquad x_0 \in \mathbb{R}$

$\lim_{x \to x_0^-} F_X(x) = F_X(x) - \mathbb{P}[X = x_0]$

c') se X ha densità allora F_X è continua

d) $0 \le F_X(x) \le 1 \qquad \forall x \in \mathbb{R}$

e) $F_X(x) = F_Y(x) \iff X \sim Y$

f) $\mathbb{P}[x_1 < X \le x_2] = F_X(x_2) - F_X(x_1) \qquad x_1 < x_2$

g) Se X ammette densità f allora

- F_X di classe C^1 su $(-\infty, x_1) \cup (x_1, x_2) \dots \cup (x_n, +\infty)$
- $F_X'(x) = f(x) \qquad x \neq x_1, \dots, x_n$
- F è continua in $x = x_1, \dots, x_n$

dove $x_1, \dots, x_n$ sono i punti di discontinuità di f

Oss. una funzione F è di classe C^1 su un intervallo aperto I se F è derivabile su I e la derivata F' è continua su I

Esempio

X con densità $f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{altrove} \end{cases}$

allora $F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \le x \le 1 \\ 1 & x \ge 1 \end{cases}$

F è derivabile in $x \neq 0$ e $x \neq 1$ e

$$F'(x) = \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 1 \\ 0 & x > 0 \end{cases}$$

Teorema

Data $F : \mathbb{R} \to \mathbb{R}$ tale che

a) $F(x_1) \le F(x_2)$ se $x_1 \le x_2$

b) $\lim_{x \to -\infty} F_X(x) = 0$ $\lim_{x \to +\infty} F_X(x) = 1$

c) F di classe C^1 su $(-\infty, x_1) \cup (x_1, x_2) \dots \cup (x_n, +\infty)$

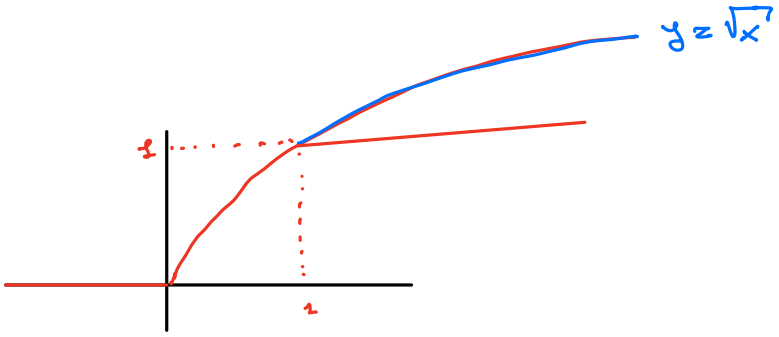
d) F è continua in $x = x_1, \dots, x_n$

allora esiste X variabile aleatoria con densità f tale che

- $F(x) = F_X(x) \quad \forall x \in \mathbb{R}$
- $f(x) = F'(x) \quad x \neq x_1, \dots, x_n$

Esempio

$$F(x) = \begin{cases} 0 & x \le 0 \\ \sqrt{x} & 0 < x < 1 \\ 1 & x \ge 1 \end{cases}$$



- F è crescente
- $F(x) \le 0$ se $x \le 0$ $\quad F(x) = 1$ $\quad x \ge 1$
- $F \in C^1$ su $(-\infty, 0) \cup (0, 1) \cup (1, +\infty)$
- F è continua in $x = 0$ e $x = 1$

$\Rightarrow$ $F = F_X$ dove X variabile aleatoria con densità

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2\sqrt{x}} & 0 < x < 1 \\ 0 & x > 0 \end{cases}$$

$f(0) = f(1) = 0$ valore arbitrario

$X \in [0, 1]$

$F(X \le x) = \sqrt{x} \qquad x \in (0, 1)$

$F(X \le x^2) = x$

$F(\sqrt{X} \le x) = x$

allora $\sqrt{X}$ ha densità uniforme

• Funzioni di variabile aleatoria

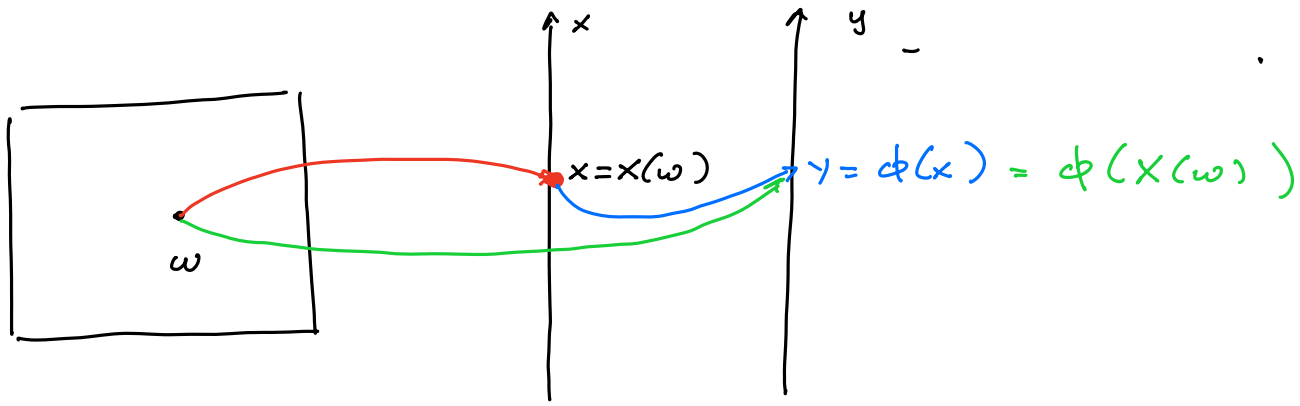
Def: $X: \Omega \to \mathbb{R}$ variabile aleatoria $x = X(\omega)$

$\phi: I \to \mathbb{R}$ funzione $y = \phi(x)$

$X \in I$

si calcola con $Y = \phi(X)$ la variabile aleatoria

$Y: \Omega \to \mathbb{R}$ $Y(\omega) = \phi(X(\omega))$



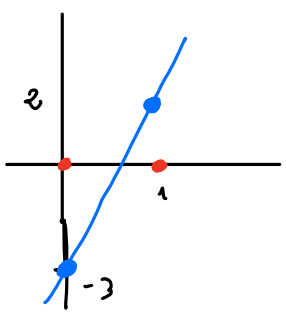
Esempio

Lancio una moneta se esce testa guadagno due euro se esce croce perdo 3 euro. Qual è il bilancio Y

$$X = \begin{cases} 1 & \text{TESTA} \\ 0 & \text{CROCE} \end{cases} \qquad Y = \begin{cases} 2 & \text{TESTA} \\ -3 & \text{CROCE} \end{cases}$$

$x = 1 \quad y = 2$

$x = 0 \quad y = -3$



$\phi(x) = mx + q = 5x - 3$

$m = \frac{2 - (-3)}{1 - 0} = 5 \qquad q = -3$

Caso discreto

$\mathrm{Im}\, X = \{ x_1, \dots, x_n, \dots \}$

$\mathrm{Im}\, Y = \{ \phi(x_1), \dots, \phi(x_n), \dots \} = \{ y_1, \dots, y_m, \dots \}$

$\# \mathrm{Im}\, Y \leq \# \mathrm{Im}\, X$ ad esempio $\phi(x) = 1$

per cui Y è una variabile discreta.

$X = \sum x_k \mathbb{1}_{E_k}$ $\quad x_i \neq x_j \Rightarrow E_i \neq E_j \Rightarrow Y = \sum \phi(x_k) \mathbb{1}_{E_k}$

Esempio: $X = \sum_{n=1}^{4} x_n \mathbb{1}_{E_n}$ $\quad x_n \neq x_m$ $\quad E_n \cap E_m = \phi$

$$Y = \phi(X) = \sum_{n=1}^{4} \phi(x_n) \mathbb{1}_{E_n} =$$

$$\phi(x_1) \neq \phi(x_2) \neq \phi(x_3) = \phi(x_4)$$
$$\| \quad\quad \| \quad\quad\quad \|$$
$$y_1 \quad\quad y_2 \quad\quad y$$

$$= y_1 \mathbb{1}_{E_1} + y_2 \mathbb{1}_{E_2} + y_3 \left(\underbrace{\mathbb{1}_{E_3} + \mathbb{1}_{E_4}}_{\mathbb{1}_{E_3 \cup E_4}} \right)$$

$$\{ \mathbb{1}_{E_3 \cup E_4} = 1 \} = \{ \phi(x) = y_3 \}$$

Legge di $Y = \phi(X)$

$$\mathbb{P}[Y = y] = \sum_{x \,|\, \phi(x) = y} \mathbb{P}[X = x] \qquad y = y_1, \dots, y_m$$

$\mathbb{P}[Y = y] = 0$ se $y \neq y_1, \dots, y_m, \dots$

Esempio: Pago 11 euro il banco lancia una moneta equilibrata 4 volte e paga il quadrato del numero di teste più 2 volte il numero di croci. Qual è la vincita netta del giocatore

X = # di teste uscite

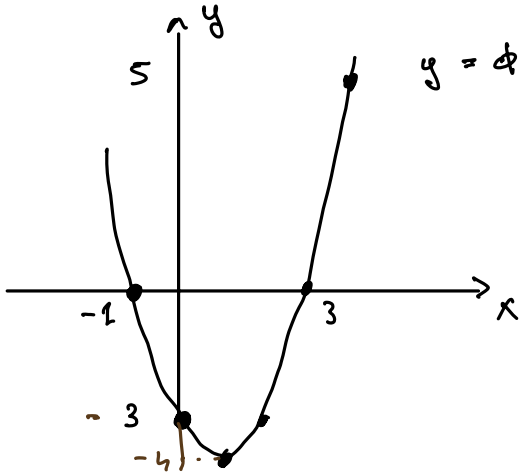
Y = vincita netta

$$Y = X^2 + 2(4 - X) - 11 = X^2 - 2X - 3$$

x	0	1	2	3	4
y	-3	-4	-3	0	5
$\mathbb{P}[X=x]$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

y	-4	-3	0	5
$\mathbb{P}[Y=y]$	$\frac{4}{16}$	$\frac{7}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

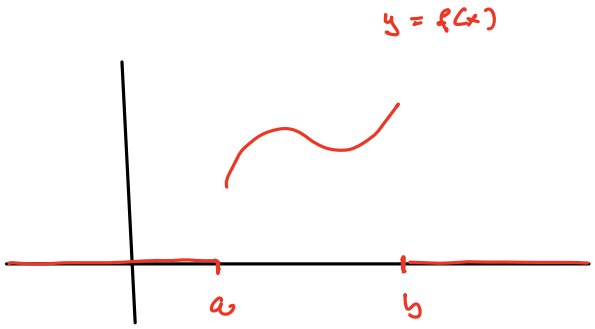
$$\phi(x) = x^2 - 2x - 3 = (x+1)(x-3)$$

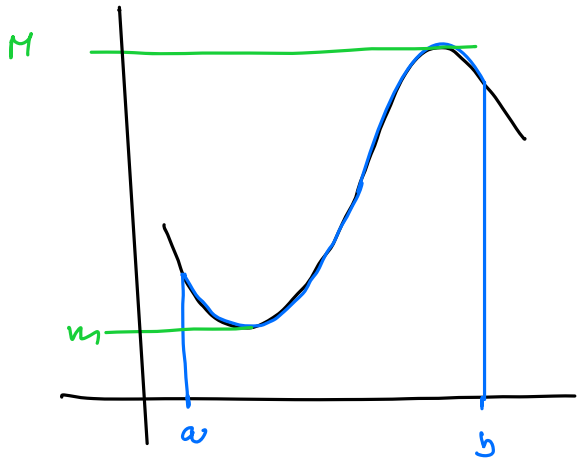


caso con densità

X ha densità $y = f_X(x)$

$Y = \phi(X)$ $\quad y = \phi(x)$

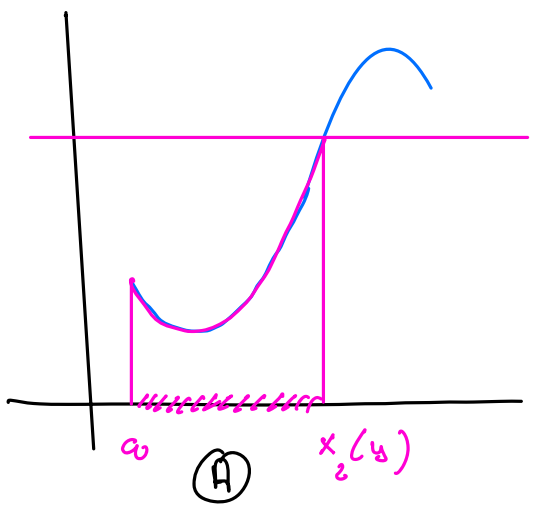




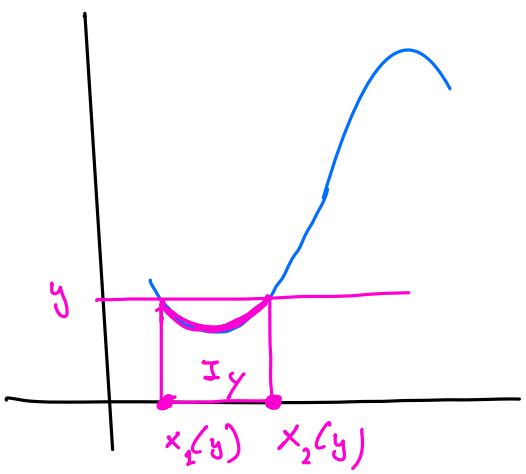
1) $a \le X \le b \quad \Rightarrow \quad m \le Y \le M \quad \Rightarrow \quad f_Y(y) = \begin{cases} 0 & y < m \\ 0 & y > M \end{cases}$

2) $m < y < M \qquad F_Y(y) = \mathbb{P}[\phi(X) \le y] = \mathbb{P}[X \in I_y]$

$I_y = \{ x \in (a,b) \mid \phi(x) \le y \} = [x_1(y), x_2(y)]$



(A)



(B)

3) se $\boxed{I_y = [a, x_2(y)]}$ allora

(A)

$$F_X(y) = F_X(x_2(y)) - F_X(a) = F_X(x_2(y))$$

poiché $F_X(a) = 0$

se $\boxed{I_y = [x_1(y), x_2(y)]}$ allora

(B)

$$F_y(y) = F(x_2(y)) - F(x_1(y))$$

4) MODO I

- calcolo $F_x(x) = \int_{-\infty}^{x} f_x(x)\,dx$

- calcolo $F_y(y) = \begin{cases} F(x_2(y)) & \text{(A)} \\ F(x_2(y)) - F(x_1(y)) & \text{(B)} \end{cases}$

deriva $f_y(y) = F_y'(y)$

4) MODO II deriva funzione composta

$$f_y(y) = F_y'(y) = F'(x_2(y))\, x_2'(y) \quad \text{(A)}$$

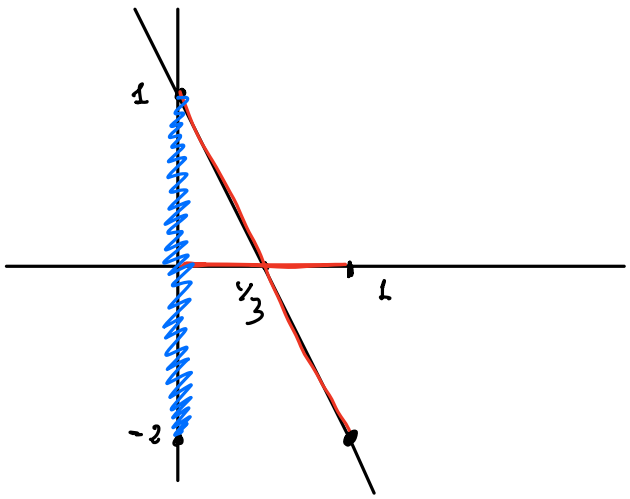
- $f_y(y) = F_y'(y) = F'(x_2(y))\, x_2'(y) - F'(x_1(y))\, x_1'(y)$

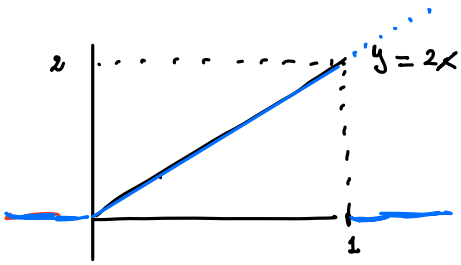
$$f_y(y) = f(x_2(y))\, x_2'(y) - F'(x_1(y))\, x_1'(y) \quad \text{(B)}$$

Lezione 18 09/04/26 2026

Esempio X ha densità $f_X(x) = \begin{cases} 0 & x < 0 \\ 2x & 0 < x < 1 \\ 1 & x > 1 \end{cases}$

$y = -3x + 1$ $\quad \varphi(x) = -x + 1$



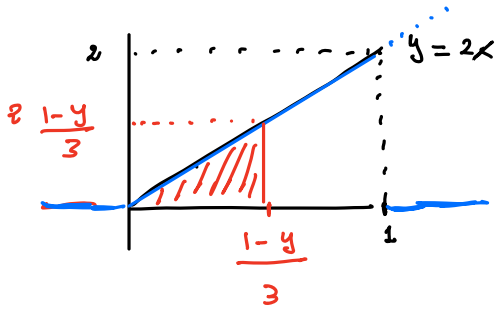


$0 \le X \le 1$ allora $-2 \le Y \le 1 \Rightarrow f_Y(y) = \begin{cases} 0 & y < -2 \\ 1 & y > 1 \end{cases}$

se $-2 < y < 1$

$$F_Y(y) = \mathbb{P}[-3X + 1 \le y] = \mathbb{P}\left[X \ge \frac{1-y}{3}\right]$$

$$= 1 - \mathbb{P}\left[X \le \frac{1-y}{3}\right] = 1 - \frac{1}{2}\left(\frac{1-y}{3}\right) 2\left(\frac{1-y}{3}\right)$$



$$= 1 - \left(\frac{1-y}{3}\right)^2 = \frac{8}{9} + \frac{2}{9}y - \frac{1}{9}y^2$$

$0 \le \frac{1-y}{3} \le 1$

se $-2 \le y \le 1$

$$\Rightarrow f_Y(y) = F_Y'(y) = \frac{2}{9}(1-y)$$

In alternativa

$$F_Y(y) = 1 - \mathbb{P}\left[X \le \frac{1-y}{3}\right] = 1 - F_X\left(\frac{1-y}{3}\right)$$

$$f_Y(y) = F_Y'(y) = \left[1 - F_X\left(\frac{1-y}{3}\right)\right]' =$$

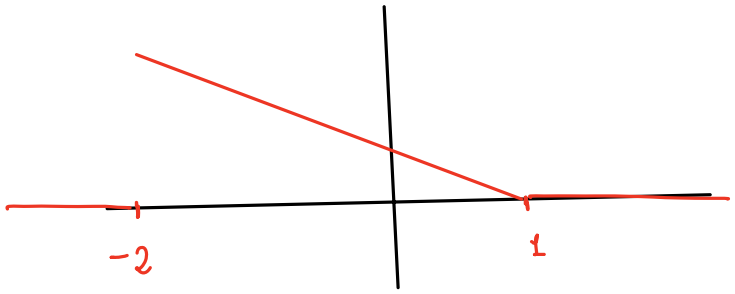
$$= -\left[F_X\left(\frac{1-y}{3}\right)\right]' = -F_X'\left(\frac{1-y}{3}\right)\left(\frac{1-y}{3}\right)' =$$

↑ DERIVATA FUNZIONE COMPOSTA

$$= -f_X\left(\frac{1-y}{3}\right)\left(-\frac{1}{3}\right) = \frac{1}{3} \cdot \frac{1-y}{3} = \frac{2}{3}(1-y)$$

Riassumendo

$$f_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{2}{9}(1-y) & -2 < y < 1 \\ 0 & y > 1 \end{cases}$$



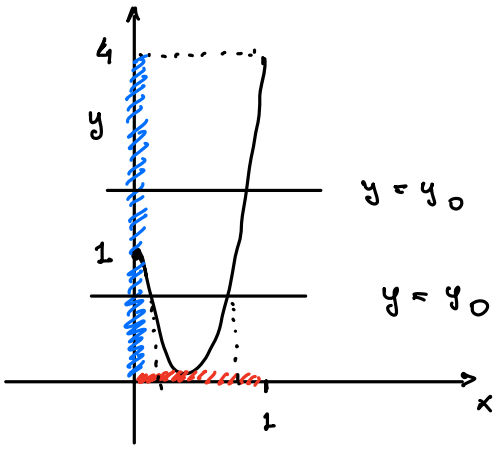
Esempio X Re densità $f(x) = \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 1 \\ 0 & x > 1 \end{cases}$

$$Y = (3X - 1)^2$$

$$\phi(x) = (3x - 1)^2$$

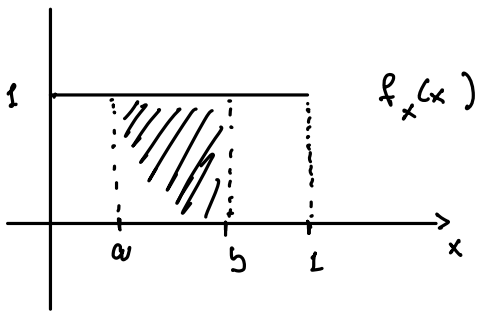
$$\phi([0,1]) = [0, 4]$$

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 < x < 1 \\ 1 & x > 1 \end{cases}$$



$$\mathbb{P}[a \leq X \leq b] = (b - a)$$

$$0 \leq a < b \leq 1$$



$$F_Y(y) = \mathbb{P}[(3X - 1)^2 \leq y] = \mathbb{P}[-\sqrt{y} \leq 3x - 1 \leq \sqrt{y}]$$

$y > 0$

$y \leq 0 \qquad F_Y(y) = 0$

$y \geq 4 \qquad F_Y(y) = 1$

$$= \mathbb{P}\left[\frac{1 - \sqrt{y}}{3} \leq X \leq \frac{1 + \sqrt{y}}{3}\right]$$

$$= \begin{cases} \dfrac{1 + \sqrt{y}}{3} & 1 \leq y \leq 4 \\ \dfrac{1 + \sqrt{y}}{3} - \dfrac{1 - \sqrt{y}}{3} & 0 \leq y \leq 1 \end{cases}$$

$$= \begin{cases} \dfrac{1 + \sqrt{y}}{3} & 1 \leq y \leq 4 \\ \dfrac{2}{3}\sqrt{y} & 0 \leq y < 1 \end{cases}$$

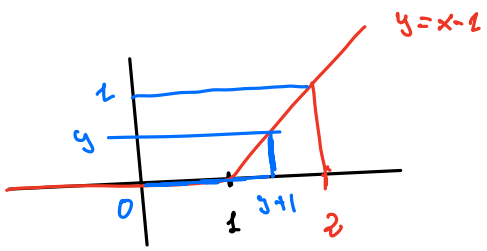
$$F_Y(y) = \begin{cases} 0 & y \le 0 \\ \frac{2}{3}\sqrt{y} & 0 < y < 1 \\ \frac{1+\sqrt{y}}{3} & 1 \le y < 4 \end{cases}$$

$$f_Y(y) = \begin{cases} 0 & y < 0 \\ \frac{1}{3}\,\frac{1}{\sqrt{y}} & 0 < y < 1 \\ \frac{1}{6}\,\frac{1}{\sqrt{y}} & 1 < y < 4 \\ 0 & y > 4 \end{cases}$$

Esempio

X ha densità $f_X(x) = \begin{cases} 1/2 & 0 < x < 2 \\ 0 & \text{altrove} \end{cases}$

$$Y = \begin{cases} 0 & X \le 1 \\ X - 1 & X \ge 1 \end{cases}$$



$0 \le X \le 2$ allora $0 \le Y \le 1$

$\mathbb{P}[Y = 0] = \mathbb{P}[X \le 1] = 1/2$ Y non ammette densità

Y non è discreta

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 1/2 & y = 0 \\ \frac{1}{2}(y+1) & 0 < y < 1 \end{cases}$$



